Solana Casino Platform — Project Plan

Competing with Juwa / Game Vault / Fire Kirin / Milky Way / Orion Stars

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1. PRODUCT VISION

Name (working): TBD — suggest something like "SolVault", "ChainCasino", "NeonReels" **Tagline:** "The First Provably Fair Crypto Casino. Every spin, every shot, every win — verifiable on-chain." **Positioning:**

- Direct competitor to Juwa/Milky Way style sweepstakes platforms
- Differentiator: **Provably fair** (VRF randomness, on-chain math, self-custody)
- No distributor layer. No sweepstakes fiction. Direct wallet play.
- Mobile-first, web3-native

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2. GAME CATALOG STRATEGY

Phase 1: Launch Catalog (MVP — 5 Games)

#	Game	Type	Why First	Complexity	Est. Dev Time
1	**SolVault**	Hold & Spin Slot	Highest revenue mechanic, proven demand, crypto-native theme	Medium	4 weeks
2	**DeepChain**	Fish Shooter	Juwa' s differentiator, multiplayer, skill-based, sticky	High	6 weeks
3	**Neon Dice**	Over/ Under Dice	Crypto casino gateway drug, instant rounds, low dev overhead	Low	1 week
4	**7 Blazing**	Classic 3-Reel Fruit	Nostalgia play, fast, mobile-perfect, low volatility	Low	2 weeks
5	**Chain Keno**	Keno	Lottery appeal, big payout potential, simple math	Low	1 week

Combined Phase 1 dev time: ~8–10 weeks (parallel tracks)

Phase 2: Expansion (Months 3–6)

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Game	Type	Why
SolLimbo	Limbo	Inverted crash, faster than crash, huge on Stake/ BC.Game
Neon Mines	Minesweeper	Grid game, extremely sticky, screenshot-sharing marketing
Chain Flip	Coin Flip	Perfect for Solana Blinks viral sharing
Dragon' s Hoard	Hold & Spin v2	Asian theme, progressive jackpot, higher volatility
Wild West Reels	5×3 Video Slot	Adventure theme, free spins, expanding wilds
Crypto Plinko	Plinko	Stake classic, simple physics, satisfying visuals

Phase 3: Premium (Months 6–12)

Game	Type	Why
Solana Blackjack	Table Game	Highest RTP game, attracts skill players
Chain Roulette	Table Game	European single-zero, 37 outcomes, classic casino staple
Tournament Fish Arena	Multiplayer Fish	4-player real-time arena, leaderboard, prize pools
NFT Slot Collection	Branded Slots	Partner with Solana NFT projects for themed reels

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3. TARGET RTP & VOLATILITY MATRIX

Game	RTP	Volatility	House Edge	Max Win	Target Hit Freq
SolVault (Hold & Spin)	96.2%	High	3.8%	2,500x	Base: 30%, Bonus: 1/120
DeepChain (Fish)	95.0%	Medium-High	5.0%	1,000x	Skill-dependent
Neon Dice	98.0%	Low	2.0%	98x (at 1.01x target)	~49%
7 Blazing (3-reel)	97.0%	Low-Medium	3.0%	500x	~25%
Chain Keno	94.0%	High	6.0%	10,000x (10/10)	Varies by picks
SolLimbo	97.0%	Adjustable	3.0%	Variable	Player-set

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Table 4 – continued

Game	RTP	Volatility	House Edge	Max Win	Target Hit Freq
Neon Mines	96.0%	Adjustable	4.0%	Variable	Player-set

Platform blended RTP target: 95.5–96.5%

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4. TECHNICAL ARCHITECTURE

4.1 Smart Contract Layer (Anchor + Rust)

```
programs/
├── house_treasury/          # Central treasury + admin controls
├── slot_engine/             # Core weighted reel randomness engine
│   ├── hold_and_spin/       # SolVault, Dragon's Hoard
│   ├── classic_reels/       # 7 Blazing, Wild West
│   └── video_slots/         # Future 5×3+ games
├── fish_arena/              # DeepChain multiplayer state
├── dice_engine/             # Neon Dice, SolLimbo
├── keno_engine/             # Chain Keno
├── mines_engine/            # Neon Mines
├── plinko_engine/           # Crypto Plinko
├── blackjack_engine/        # Future table games
├── vrf_integration/         # ORAO VRF wrapper + callback handlers
└── player_session/          # Player balances, bet history, achievements
```

4.2 Key PDAs (Program Derived Addresses)

PDA	Purpose	Data Stored
`GameConfig`	Immutable game parameters	RTP, paytable, reel weights, max bet, min bet
`HouseTreasury`	Platform bankroll	USDC balance, SOL reserve, admin pubkey, pause flag
`PlayerSession`	Per-player state	Current USDC balance, total wagered, total won, active bet
`PendingSpin`	VRF request in flight	Player, bet amount, game type, request timestamp, force seed
`BonusRound`	Hold & Spin active bonus	Locked symbols[], respins remaining, round seed, total value
`FishArena`	Multiplayer fish game state	Fish[] (type, pos, HP, value), players[], timestamp, round ID
`BetHistory`	Audit trail	All bets, outcomes, payouts (for provably fair verification)

4.3 VRF Integration (ORAO)

```
Player clicks SPIN
```

```
    → Client calls `place_bet()` instruction
    → Program: deduct USDC from player → store in PendingSpin PDA
    → Program: CPI to ORAO VRF `request()` with callback to `resolve_spin()`

ORAO Oracle fulfills (off-chain computation + on-chain proof)
    → ORAO program verifies proof
    → ORAO program calls `resolve_spin()` callback into our program
    → Our program: verify VRF result matches pending request
    → Our program: map random bytes to reel positions using weighted cumulative distribution
    → Our program: evaluate win, calculate payout
    → Our program: if win → transfer from HouseTreasury to player
    → Our program: if bonus trigger → init BonusRound PDA
    → Our program: close PendingSpin PDA
    → Our program: emit event for frontend
```

4.4 Frontend Stack

Layer	Tech	Why
Framework	Next.js 14 (App Router)	SSR, API routes, Vercel deployment
Language	TypeScript	Type safety across full stack
Styling	Tailwind CSS + shadcn/ui	Fast UI development, mobile-first
Wallet	@solana/wallet-adapter	Phantom, Solflare, Backpack, Glow
On-chain	@coral-xyz/anchor	Auto-generated TypeScript clients from IDL
RPC	Helius / QuickNode	Fast, reliable, websocket support for real-time updates
Game Engine	Phaser 3 (fish/table) + React (slots/dice)	Phaser for real-time fish arena, React for UI-heavy games
State	Zustand + React Query	Client state + server/cache state
Animations	Framer Motion + GSAP	Smooth reel spins, bonus round sequences
Sounds	Howler.js	Web audio, sprite sheets for SFX

4.5 Backend / Indexer

Component	Tech	Purpose
Indexer	Helius webhooks / custom Geyser	Real-time on-chain event ingestion

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Table 7 – continued

Component	Tech	Purpose
API	Next.js API routes / tRPC	Player stats, leaderboard, game history
Database	PostgreSQL (Supabase)	Player profiles, aggregated stats, off-chain config
Cache	Redis (Upstash)	Session state, rate limiting, hot data
Auth	Wallet-based (SIWS)	Sign-in with Solana, no passwords

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5. TOKENOMICS & ECONOMY

5.1 Betting Currency: USDC (SPL)

- **Primary play currency**: USDC (SPL Token)
- **Why not SOL?** Price stability during sessions. Players hate seeing their stack fluctuate 5% because SOL moved.
- **Deposits**: Direct USDC transfer to PlayerSession PDA
- **Withdrawals**: Program transfers from HouseTreasury to player wallet
- **Gas**: SOL for transaction fees (minimal, ~$0.00025/tx)

5.2 Platform Token (Optional Phase 2)

Token Name: $VAULT (or $SPIN, $NEON) **Standard:** SPL Token or Token-2022 with transfer hooks

Utility	Mechanic
Staking	Stake $VAULT → earn share of house revenue (weekly USDC airdrop)
Fee discount	Hold $VAULT → reduced house edge (e.g., 96.5% → 97.5% RTP)
Exclusive games	High-stakes rooms, NFT slots, tournament entry
Burn mechanism	0.5% of every bet burned → deflationary
Airdrops	Retroactive to early players based on volume

5.3 House Treasury Management

```
House Treasury PDA
├── USDC reserve (payouts)
├── SOL reserve (gas, oracle fees)
├── Platform revenue wallet (70% of house edge)
├── Staking rewards wallet (20% of house edge)
├── Development fund (10% of house edge)
└── Insurance buffer (minimum 500x max bet)
```

Bankroll math:

- Max bet = 1% of treasury (prevents ruin)

- Insurance buffer = 500 × max bet = 5× treasury
- If treasury drops below insurance threshold → automatic bet limit reduction

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6. DEVELOPMENT ROADMAP

Month 1: Foundation

Week	Deliverable
1	Anchor project scaffold, HouseTreasury program, PlayerSession PDA, basic admin controls
2	VRF integration (ORAO), test on devnet, callback pattern solidified
3	Dice engine (Neon Dice) —simplest game, end-to-end flow: bet → VRF → resolve → payout
4	Dice frontend, wallet connect, basic UI, deploy to devnet, internal testing

Month 1 Goal: One working game (Dice) on devnet, full stack proven.

Month 2: Core Games

Week	Deliverable
5	Slot engine (weighted reel randomness), 3-reel classic (7 Blazing)
6	7 Blazing frontend, reel animation, payline evaluation, mobile optimization
7	Hold & Spin engine (SolVault), BonusRound PDA, respin logic, special symbols
8	SolVault frontend, bonus round UI, jackpot animations, sound design

Month 2 Goal: Two slot games (classic + Hold & Spin) + Dice on devnet.

Month 3: Fish Arena + Polish

Week	Deliverable
9	Fish arena program: Arena PDA, fish spawn logic, HP system, shot validation
10	Fish arena Phaser.js client, multiplayer sync, weapon system, boss battles
11	Keno engine (Chain Keno), frontend, number selection UI
12	Security audit (OtterSec/Neodyme), bug fixes, testnet stress test, mainnet prep

Month 3 Goal: 5 games on mainnet-beta, security audited, 100 beta testers.

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Month 4: Launch + Growth

Week	Deliverable
13	Mainnet launch (5 games), marketing site, docs, provably fair verification page
14	Referral program, affiliate dashboard, social sharing (Blinks for Coin Flip)
15	Leaderboards, player stats, achievements system
16	Limbo + Mines expansion, A/B testing on game order/promotions

Months 5–6: Scale

- Blackjack, Roulette table games
- Tournament system (fish arena leaderboards, slot races)
- Mobile app (React Native wrapper or PWA)
- $VAULT token launch (if Phase 2)
- Multi-language support (Spanish, Portuguese, Vietnamese)

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7. PROVABLY FAIR SYSTEM

This is your #1 marketing asset. Build it like a product feature.

7.1 Public Verification Page

```
/provably-fair
├── Game: [Dropdown]
├── Reel Weights: [JSON download]
├── Paytable: [Visual table]
├── RTP Calculator: [Simulate 1M spins]
├── VRF Contract: [ORAO address + link]
├── House Treasury: [Real-time USDC balance]
└── Recent Spins: [Table with tx signatures, VRF proofs, outcomes]
```

7.2 On-Chain Transparency

Data	Location	Immutable?
Reel symbol weights	GameConfig PDA	Yes (init-only)
Paytable	GameConfig PDA	Yes (init-only)
RTP	GameConfig PDA	Yes (init-only)
VRF results	Transaction logs	Yes (on-chain)
Bet history	BetHistory PDA + indexer	Yes (append-only)
Treasury balance	HouseTreasury PDA	Real-time

7.3 Player-Facing Proof

Every spin result shows:

- Transaction signature (Solana Explorer link)
- VRF request ID + proof hash
- Random bytes generated
- How those bytes mapped to reel positions
- Payout calculation step-by-step

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8. LEGAL & COMPLIANCE STRATEGY

8.1 Jurisdictional Approach

DO NOT:

- Target US players with real-money gambling without state licenses
- Operate as an unlicensed casino in regulated markets
- Allow players from prohibited jurisdictions

DO:

- Geoblock US, UK, Australia, and other regulated markets at launch
- Target crypto-native players in permissive jurisdictions (LatAm, SE Asia, Eastern Europe, Africa)
- Implement KYC for withdrawals above $2,000 equivalent
- Maintain AML compliance (Chainalysis/Syndicate for wallet screening)

8.2 Licensing Path (Long-term)

Jurisdiction	Cost	Timeline	Notes
Curaçao eGaming	~$20–40K	2–3 months	Cheapest entry, less prestige
Anjouan (Comoros)	~$15–25K	1–2 months	Emerging, crypto-friendly
Isle of Man	~$100K+	6–12 months	Gold standard, crypto-friendly
Malta MGA	~$150K+	12–18 months	EU credibility, strict requirements

Recommendation: Start unlicensed in permissive markets, pursue Curaçao or Anjouan within 6 months for credibility.

8.3 Terms of Service Structure

- Platform is “entertainment” — players wager USDC against smart contracts
- No guarantees of winning
- Self-exclusion tools (cool-off periods, permanent bans)
- Responsible gambling messaging

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- Dispute resolution via on-chain arbitration (optional)

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9. GO-TO-MARKET STRATEGY

9.1 Launch Strategy: "Fairness First"

Narrative: "Every other casino hides the math. We put it on-chain."

Launch Channels:

1. **Crypto Twitter / X** — Thread series on provably fair mechanics, VRF deep dives
2. **Discord community** — Beta testers, bug bounty, feedback loop
3. **Solana ecosystem** — Partner with Phantom, Solflare, Jupiter for co-marketing
4. **Influencer seeding** — Crypto gambling streamers (Kick, Twitch) with small bankrolls
5. **Referral program** — 1% of referred player's house edge → referrer (in USDC)

9.2 Player Acquisition Tactics

Tactic	Mechanic	Cost
Faucet / Free Play	New wallet gets $0.50 USDC to try (funded by platform)	Low
Daily Spin Wheel	Log in daily → spin for bonus USDC or free spins	Retention
Tournaments	Weekly slot races, fish arena battles → prize pools	Marketing
Affiliate Program	Distributors/agents earn 20–30% of house edge from referred players	Performance
Blink Sharing	Share a Coin Flip link → friend plays from Twitter/Discord directly	Viral

9.3 Retention Mechanics

- **Streak bonuses**: Consecutive daily logins → escalating rewards
- **VIP Tiers**: Bronze/Silver/Gold/Platinum based on lifetime wager volume → perks, exclusive games, faster withdrawals
- **Achievement system**: "First 100x win", "10,000 spins", "Fish Slayer" → NFT badges
- **Community events**: Scheduled fish arena boss battles with guaranteed prize pools

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10. TEAM STRUCTURE (Recommended)

Role	Headcount	When Needed
Founder / PM	1 (you)	Day 1
Rust/Anchor Dev	1–2	Day 1

Table 16 – continued

Role	Headcount	When Needed
Frontend Dev (React/ Next.js)	1–2	Day 1
Game Dev (Phaser/ Canvas)	1	Month 2 (fish arena)
UI/UX Designer	1	Week 2
Sound Designer	1 (contract)	Month 2
Community Manager	1	Month 3
Security Auditor	1 (contract)	Month 3
Legal Counsel	1 (advisory)	Month 1

Minimum viable team for launch: 4 people (1 Anchor dev, 1 frontend dev, 1 designer, 1 PM/ founder)

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11. BUDGET ESTIMATE (6-Month Runway)

Category	Cost (USD)	Notes
Development	$60,000–120,000	2 devs × 6 months at $5–10K/mo each
Design	$15,000–25,000	UI/UX + game art + sound
Security Audit	$15,000–30,000	OtterSec/Neodyme for 5 programs
Infrastructure	$5,000–10,000	RPC (Helius), hosting (Vercel), DB (Supabase), Redis
VRF / Oracle Costs	$2,000–5,000	Per-request fees at scale
Marketing / Launch	$20,000–50,000	Influencers, ads, community building
Legal / Compliance	$10,000–20,000	Counsel, terms, geoblocking, KYC integration
House Bankroll	$50,000–100,000	Minimum treasury to cover early payouts
Buffer	$20,000–40,000	Unexpected costs
TOTAL	**$197,000–400,000**	

Lean version (2 devs, minimal marketing, smaller bankroll): ~$100,000

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12. SUCCESS METRICS

Metric	Month 3 Target	Month 6 Target	Month 12 Target
DAU	100	1,000	5,000
Monthly Wager Volume	$50,000	$500,000	$5,000,000
House Revenue (5% edge)	$2,500	$25,000	$250,000
Games Live	5	8	12+
Avg Session Length	10 min	15 min	20 min
Player Retention (D7)	20%	30%	35%
Security Incidents	0	0	0

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13. RISK REGISTER

Risk	Likelihood	Impact	Mitigation
Smart contract exploit	Medium	Critical	Multiple audits, bug bounty, insurance (InsurAce/Nexus Mutual)
VRF oracle failure	Low	High	Fallback oracle (MagicBlock), manual resolution protocol
Regulatory shutdown	Medium	Critical	Geoblocking, no US marketing, KYC, legal counsel
House bankroll drain	Medium	High	Bet limits, insurance buffer, volatility monitoring
Competitor copy	High	Medium	First-mover advantage, brand, community, continuous innovation
Low player acquisition	Medium	High	Strong GTM, referral program, free play faucet, Blinks virality
Solana network issues	Low	Medium	Multi-RPC fallback, transaction retry logic

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14. NEXT STEPS (This Week)

1. **Finalize game name** — Brand identity, domain, social handles
2. **Set up dev environment** — Anchor, Rust, Solana CLI, ORAO SDK
3. **Create GitHub repo** — Monorepo: `programs/`, `app/`, `sdk/`
4. **Write first program** — `house_treasury` + `player_session` PDAs
5. **Integrate ORAO VRF on devnet** — Get a working `request → callback` flow
6. **Build Neon Dice** — Your first end-to-end game (1 week max)
7. **Create landing page** — "Coming soon" with waitlist, provably fair manifesto

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8. **Join Solana dev communities** — SuperteamDAO, Solana Tech Discord, find co-founders

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APPENDIX A: RECOMMENDED GAME NAMES

Name	Theme	Available?
SolVault	Vault/heist	Likely available
NeonReels	Cyberpunk slots	Likely available
ChainSpin	Generic crypto	Likely available
DeepChain	Underwater/fish	Likely available
BlockBets	Blockchain casino	Likely available
CryptoCade	Arcade crypto	Likely available
OnChain Casino	Literal	Likely available
FairSpin	Provably fair	Likely available

APPENDIX B: COMPETITIVE MATRIX

Feature	Juwa	Milky Way	Fire Kirin	Your Platform
Provably Fair	❌	❌	❌	✅ On-chain VRF
Self-Custody	❌	❌	❌	✅ Wallet holds funds
Transparent RTP	❌	❌	❌	✅ Immutable on-chain
No Distributor	❌	❌	❌	✅ Direct play
Mobile App	✅	✅	✅	✅ Web3 mobile
Fish Games	✅	✅	✅	✅
Slots	✅	✅	✅	✅
Instant Payouts	⚠️ (agent)	⚠️ (agent)	⚠️ (agent)	✅ Smart contract
Crypto Native	❌	❌	❌	✅ Built on Solana
Community Ownership	❌	❌	❌	✅ $VAULT token (Phase 2)

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